The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

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MAT 3007 — Optimization

Exercise Sheet 7

Problem 1 (Convex Set): (*20* pts)
Are the following sets convex? Explain your answer.

a) $S = \{s_1 + s_2 : s_1 \in S_1, s_2 \in S_2\}$, where S_1 and S_2 are convex.

b) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 5\}$

c) $\{x \in \mathbb{R} : ax^2 + 2x - \frac{1}{a} \geq 0\}$ with $a > 0$.

d) $\{(x, y) \in \mathbb{R}^2 : y \geq |x|\}$.

e) $\{\boldsymbol{x} \in \mathbb{R}^2 : \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \leq 0\}$ where $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Problem 2 (Convex Function): (*20* pts)
Are the following functions convex? Explain your answer.

a) $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_1 = \sum_{i=1}^d |x_i|$.

b) $f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{y}\|^2$, where $\boldsymbol{y}$ is a given vector in $\mathbb{R}^d$.

c) $f : \mathbb{R}^d \to \mathbb{R}$ satisfying $f(\lambda \boldsymbol{x}) = |\lambda| f(\boldsymbol{x})$ and $f(\boldsymbol{x} + \boldsymbol{y}) \leq f(\boldsymbol{x}) + f(\boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$ and any $\lambda \in \mathbb{R}$.

d) $f = |x|^{3/2}$.

e) $f(x) = \ln(x + \sqrt{1 + x^2}) + x^2$.

Problem 3 (True or False): (*20* pts)
For each of the following statements, determine whether it is true or false. Provide a justification for your answer.

a) The union of a finite number of convex sets is always convex.

b) If f and g are convex functions, then the composite function $h = f \circ g$, i.e., $h(\boldsymbol{x}) = f(g(\boldsymbol{x}))$, is always a convex function.

c) If f and g are convex functions, then the function $h = f \cdot g$, i.e., $h(\boldsymbol{x}) = f(\boldsymbol{x}) \cdot g(\boldsymbol{x})$, is always a convex function.

d) If f is a convex function, then the set $S = \{(y, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n \mid y = f(\mathbf{x})\}$ is a convex set.

e) A line segment in $\mathbb{R}^2$ is always a convex set.

Problem 4 (Global Optimality Condition): (*20* pts)
Suppose that we have centered observations $(\boldsymbol{x}_i, y_i)$ such that $\sum_{i=1}^n \boldsymbol{x}_i = \mathbf{0}, \sum_{i=1}^n y_i = 0$. Let $(\beta_0^\star, \boldsymbol{\beta}^\star)$ be a global minimizer of the least squares objective function

$$f(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} (\beta_0 + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i)^2.$$

a) Prove that $\beta_0^\star = 0$.

b) The observations before centering are $(\boldsymbol{x}_i', y_i')$ and $\boldsymbol{x}_i' = \boldsymbol{x}_i + \boldsymbol{q}$, $y_i' = y_i + r$ for all i. Show that $(\beta_0, \boldsymbol{\beta})$ minimizes f if and only if $(\beta_0 - \boldsymbol{\beta}^\top \boldsymbol{q} + r, \boldsymbol{\beta})$ minimizes

$$\tilde{f}(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} (\beta_0 + \boldsymbol{\beta}^T \boldsymbol{x}_i' - y_i')^2.$$

Problem 5 (KKT Condition for Convex Optimization): (*20* pts)
Consider the following optimization problem

$$\begin{array}{ll} \min_{x_1, x_2} & x_1^2 + x_2^2 \\ \text{subject to} & x_1 + 2x_2 \leq -3 \end{array}$$

Answer the following questions:

a) Find all all KKT-points.

b) Determine which of these points are global minimizers. Justify your answer.

c) Are there any other global minimizers besides the ones identified in part (2)? Explain why or why not.

Sheet 7 is due on **April 21st**. Submit your solutions before **April 21st**, **11:00 pm**.